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A Novel Predictive Analysis and Classification of Land Subsidence Vulnerability Mapping based on GIS using Hybrid Optimized Machine Learning Techniques and Computer Vision

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Abstract

Land subsidence is a natural disaster caused by the extraction of the earth's material strength on the surface that causes the earth's surface to settle and sink. Land subsidence can also result in significant financial losses since it causes structural damage and expensive maintenance costs for roads, trains, barriers, pipelines, ground water, oil, natural gas, mines, and buildings. The Geographic Information System (GIS) and Google Earth Platform (GEP) can be used to map this land subsidence. This provides a comprehensive view of land pattern subsidence and its interactions with other geographical elements. It applies to both progressive subsidence caused by fluid extraction (groundwater, oil, or natural gas) and rapid surface subsidence caused by the collapse of an underground mine. Risk mitigation is the most effective way to deal with the experimental model. This research study focuses on the use of Predictive Data Analysis (PDA), a powerful environment in data analytics that provides a set of tools that make it easier to evaluate optimal models. The proposed model makes use of a dataset that includes specific land subsidence circumstances as input. The prediction result is achieved using a hybrid machine learning technique that combines Stochastic Gradient Descent (SGD) with Long Short-Term Memory (LSTM) and is compared to K-Nearest Neighbor (KNN), Generative Adversarial Network (GAN), and Artificial Neural Network (ANN). The proposed system employs LSTM and SGD as optimizers, resulting in a hybrid model that improves accuracy. Among the four algorithms investigated the LSTM-SGD network has the highest accuracy (97%), followed by KNN (96%), GAN (90%), and ANN (89%). Due to their ability to capture complicated nonlinear correlations in data and manage temporal dependencies, optimized hybrid ML models LSTM-SGD and KNN models appear to be particularly well-suited for land subsidence prediction.

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Keywords: Land Subsidence, Predictive Data Analysis (PDA), Geographic Information System, Machine Learning, Computer Cision, Susceptibility Mapping, Stochastic Gradient Descent (SGD), Generative Adversarial Network (GAN), Long Short-Term Memory (LSTM), K-Nearest Neighbor (KNN), Artificial Neural Network (ANN), Google Earth Platform.

1. Introduction

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The practice of making maps that depict the regions of the earth that have sunk over time is known as land subsidence mapping. This data is critical for understanding the extent and effect of land subsidence, which can have serious implications on infrastructure, ecosystems, and human populations. Land subsidence maps may be used to detect locations prone to subsidence, which can aid in infrastructure planning and development choices [1]. It can track the amount and pace of ground sinking over time. This data can be used to pinpoint regions where mitigating actions are required. These can be used to detect the variables causing land subsidence in a specific location. Team led by ecologist Madhav Gadgil assessed severe ecological degradation in the Western Ghats and classified 64% of the territory as highest sensitive, highly sensitive, or moderate sensitive.

This data may be used to build mitigation methods for land subsidence. Facilities, environments, and urban centers can all suffer as a result of land subsidence. It effects of natural disaster land subsidence can cause structural damage to bridges and highways. It may also cause saltwater to spill into water sources and wetlands. In severe cases, land subsidence can cause entire structures to collapse. It is a developing problem all across the world, and it is critical to establish effective solutions for dealing with it. Land subsidence charting is an important technique for determining the size and effects of subsidence of land and creating mitigation and adaptation methods [2]. Finally, the results obtained from PDA were digitized in an ArcGIS software and exported into .kml file for better interpretation and visualization of results in Google Earth platform and Prediction Scatter Plot Over a Map for land subsidence as shown in figure 1 and 2.

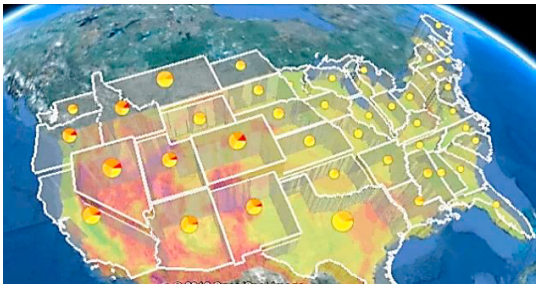


Fig. 1. Geographic Landform Data in an Earth Browser KML files to pinpoint locations

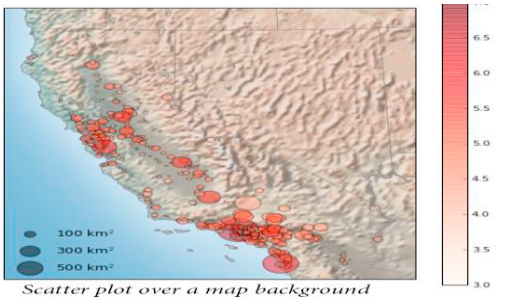


Fig. 2. Prediction Scatter Plot Over a Map for Land Subsidence

GIS allows for a seamless combination of various sources of data, including satellite imaging, ground-based tests, and geologic information, into a cohesive geographic framework. This provides a comprehensive perspective of the subsidence of land patterns and their interactions with other geographical factors. It provides spatial analytic methods, including buffer evaluation, distance assessment, and interpolation methods, for finding locations susceptible to land subsidence and analyzing the possible implications on nearby infrastructure and ecosystems [3].

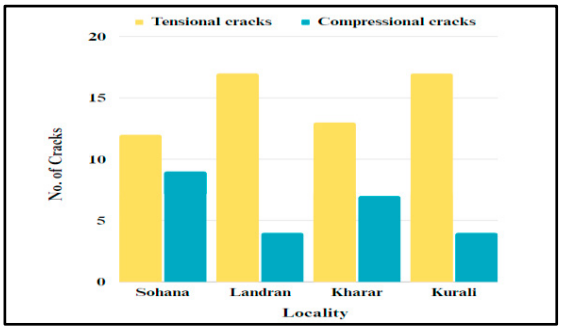


Fig. 3. No of Cracks in various Indian Localities

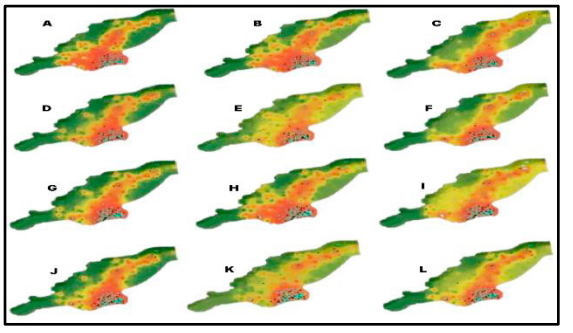


Fig. 4. Land Subsidence Mapping under different Conditions

This is critical for tracking land subsidence since the elements that lead to sinking might vary as time passes. Creating revolutionary machine learning techniques built exclusively for land subsidence prediction. The

application of group learning approaches to mixing different methods of machine learning towards greater accuracy is being investigated. Figure 3 gives the number of Cracks recorded in various Indian Localities, with which we can see that Tensional and compressional crack over a wide range of regions on the Indian Locality. Figure 4 shows the land subsidence mapping under various conditions though GIS [4]. The proposed system employs the use of machine learning and GIS to use the tracking of land subsidence mapping. The factors of these mapping are considered for the prediction process which is initiated by preprocessing and carried out using a hybrid model using ANN, GAN, KNN, and LSTM-SGD. The efficiency and accuracy evaluation results are analyzed to estimate the optimization hybrid algorithms.

2. Related Work

D. Yuan et. al [5] in their work demonstrated a system with the highest discrepancy among InSAR scheduling analysis outcomes with GPS tracking information over the same period is 10.91mm, demonstrating scheduling stack with InSAR methodology is highly trustworthy in land degradation. The increased excess correction and fluid allocation in the proposed model enhances efficiency score when it is made as comparison with Grey-Markov model. This simulation results with optimized remnant reduction and dynamic empowerment are better appropriate for the real variation of the problem in the intensity of damage region.

Y. He et. al [6], suggested CNN-PhLSTM model has been contrasted with the SVR and MLP, and CNN-LSTM models using SSIM can be used with MAE. The proposed model using these is proved to be efficient and accurate. The findings indicate three locations of continuing sinking, including some regions of Longhsou, western, and eastern excavation areas. From 2014 to 2021, overall surface deformation increased. The geographic location of exterior strain within the Jinchuan mining region is controlled by faults and lithology. The prediction findings reveal overall the outermost deformation spectrum will continue to grow whereas time-series area deformation will slow down in the following two years.

L. Zhang et. al [7] proposed a system on elemental basis approach, strain observation, time line prediction, the research developed an immediate breakdown warning system for gearbox towers underground subsidence conditions. The missing points on the development elevated spot were discovered by modeling the load placement of the structure in actual foundation movements. Deploying a real-time stress tracking for developing the ARIMA timeline prediction system allows for early signal based on strain monitoring and forecasting data. The approach suggested in this work is efficient and cost friendly and it may be utilized to undertake emergency treatment of risky tower underground subsidence conditions.

B. Hu et. al [8] in their work, with SAR picture extract data for offset handling method to combine InSAR using terrestrial 3D extraction methodology. These technical characteristics, evaluation outcome, implementations, along with limits of these approaches, addressed in this study, and some prospective possibilities of employing SAR pictures to evaluate quick and significant subsidence in excavation sites can be explored at last.

3. Proposed work

3.1 Predictive Data Analytics

Predictive data analysis, frequently referred to as forecasting, is a method of identifying the probability of upcoming occurrences using previous data utilizing information, data mining techniques, advance machine learning algorithms, and statistical approaches. Predictive analytics are used to discover challenges along with opportunities in this data. This entails analyzing data structures, patterns, and linkages in order to forecast potential events or behaviors. It is critical in predicting land subsidence, that is the progressive leveling or lowering of the earth's surface. It assists in the development of credible models for anticipating future subsidence patterns and limiting potential dangers by utilizing diverse data sources and modern analytical approaches. PDA can be utilized to track and forecast land subsidence induced by significant extraction of groundwater, allowing for the creation of long-term water management policies. It aids in mapping and forecasting subsidence resulting from oil and gas

manufacturing, easing permitting procedures, and preserving essential facilities. Implementation using the python codes are used for the predictive data analysis of the big data.

4. Methodology

Satellite photography, ground-based measurements, and LiDAR (Light Detection and Ranging) data are all used to acquire land subsidence data [9]. The information gathered is used to construct subsidence maps, which indicate the regions of land that have sunk.[10] The subsidence maps are used to generate a land subsidence dataset. There are three phases to the dataset: training, testing, and validation. The machine learning hybrid model including LSTM-SGD, KNN, GAN, and ANN can learn long-term relationships in data as shown in Fig.5, allowing it to forecast subsidence in new locations with high accuracy. As a result, a model is a useful tool for tracking land subsidence and analyzing its impact on infrastructure and ecosystems [11].

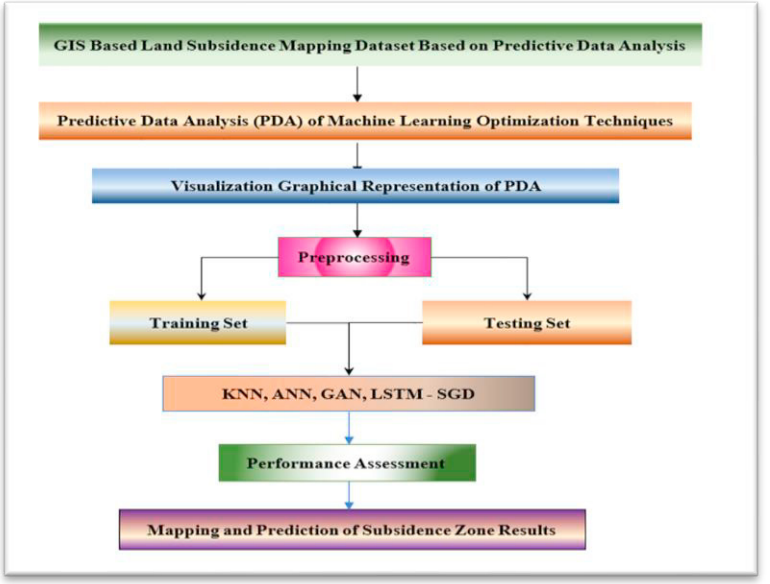


Fig. 5. Proposed Architecture Diagram

4.1 Dataset Description

The image is a visualization of a dataset of land subsidence data that was collected from various sources, including satellite imagery, ground-based measurements, and LiDAR (Light Detection and Ranging) data. The dataset was used to train a machine learning model, which is a type of artificial neural network that can learn long-term dependencies in data. The LSTM model was able to accurately predict the subsidence in new areas, and it is a valuable tool for monitoring land subsidence and assessing its impact on infrastructure and ecosystems. The collection contains measurements of vertical ground surface displacement from over 200 of the most heavily used and inhabited groundwater basins. Conditions Affecting Land Subsidence are shown in table 1.

Table. 1 Conditions Affecting Land Subsidence

Conditions for Land Subsidence	Percentage Affecting Zone (in %)
Elevation	82
Aspect	68
Closeness to road	63
Earthquake	52
Fault removal	12
TWI	23
Soil Erosion	10
Rainfall	5
Mineral extraction	14

Slope	20
Underground water pumping	33
Land use	56

4.2 Artificial Neural Network (ANN)

ANNs are a form of computational learning algorithm that is influenced on the working of anatomy and characteristics of the human brain. These can be made up of using linked nodes known as neurons, which interact with one another by transmitting signals. The intensity of a signal impacts its influence on the receiving neuron, and it is weighted. Neurons are arranged in layers, and neuronal connections may be both forward-feeding or recurrent. It undergoes training on data by altering the weights of neural connections. The weights of the ANN's connections are then adjusted until they can provide the desired result for each input. They are a sort of neural network algorithm that has been demonstrated to help forecast land subsidence. It can learn complicated correlations between input and output data and may be used to predict a wide range of land subsidence parameters such as groundwater extraction, geological conditions, and land usage. They have the potential to anticipate land subsidence. They can learn intricate correlations between data inputs and outputs, and they can simulate a wide range of land subsidence causes. These have been demonstrated in several studies to be successful in predicting land subsidence, and they can be utilized to construct early detection systems for land subsidence.

$$z = \sum_{i=n}^{n}(w_i \,.\, x_i) + b \tag{1}$$

where z is the input to the neuron, wi are the weights, xi is the inputs, b is the bias, and n is the number of inputs. The activation function is a function that is not linear and generates an output from the weighted sum.

$$f(z) = \frac{1}{1+e_{-z}} \tag{2}$$

$$E = \sum(y_i - y\hat{}i)(y_i - y\hat{}i)^2 \tag{3}$$

where f(z) is the output of the activation function, z is the weighted sum E is the error, y_i is the actual output, and $\hat{y}_i$ is the predicted output. Fig. 6. shows the working architecture of ANN model.

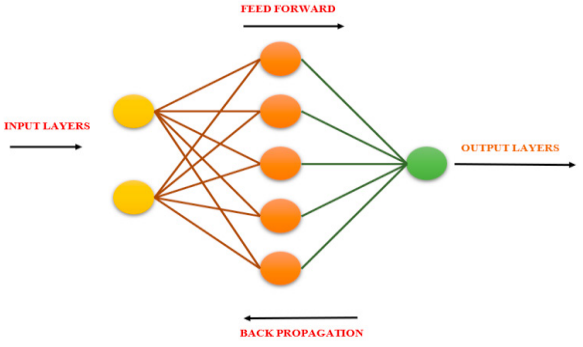


Fig. 6. ANN - Working Model

4.3 Generative Adversarial Networks (GAN)

GANs (Generative Adversarial Networks) are a form of a machine learning system that generates novel data that is comparable to current data. GANs are made up of a pair of neural networks, one for the machine that generates and one for the discriminator. The generator must create new data, whereas the discriminator's goal is to tell the difference between actual and false data. Both the discriminator and generator are adversarially trained, which means they are always attempting to outwit each other. The generator is attempting to produce phony data that is sufficiently convincing that the discriminator cannot detect the difference between it and actual data. The algorithm that discriminates is attempting to improve its ability to distinguish between actual and bogus data to detect the generator's trickery. GANs (Generative Adversarial Networks) have emerged as a potential tool for predicting land subsidence, with significant benefits over older approaches. Additionally, these approaches may

fail to capture complicated interactions among land subsidence and numerous contributing factors, resulting in less accurate projections. Improve the precision and resilience of GAN-based prediction models. Comprehensible GAN models to learn more about land subsidence mechanisms to Combine GANs alongside additional machine learning techniques to forecast and analyze land subsidence.

$$L_G = -Ex[(Log\ Dx)] - Ez[\log(1 - D(Gz))] \quad (4)$$

$$V(D,G) = Ex - p(x)[(Log\ D(x)] + Ez - p(z)[\log(1 - D(Gz))] \quad (5)$$

The generator loss is a measure of how well the generator can fool the discriminator. It is typically calculated using the cross-entropy loss function. The discriminator loss is a measure of how well the discriminator can distinguish between real and fake data. It is also calculated using the cross-entropy loss function. The minimax game approach is used to train the GAN utilizing the generator and discriminator losses. The minimax game's purpose is to minimize the generator loss while maximizing the discriminator loss. This antagonistic training process pushes both the generator and the discriminator to get better over time.

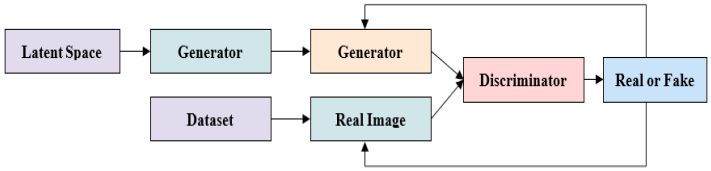


Fig. 7. GAN Architecture

4.4 K-Nearest Neighbour (KNN)

Consideration of classification (clustering) in this type of experiment to explore a common ground of difference through the distance between each point, which segregates minute differences and classifies them in their coordinate points of 'x' and 'y'. KNN classification simplifies the analysis of robust data and provides considerable results in pattern identification via binomial or multi-class classification. The major determinant is the K-value and the distance between the locations. The normalization of the dataset makes a difference in the model's performance, which is already finished in the EDA process by determining the 'K' value. In the default situation, it estimates the middle point between two distinct regions, such as Euclidean distance, however, this may be changed by hyper-parameter adjustment.

$$d(x,y) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2} \quad (6)$$

The KNN models differ in their use of distance-finding formulas; there are three types of distance-finding formulas: Euclidean, Cartesian, and Mercator. Distance: This parameter allows us to measure an object's position shift from one state to another. This indicates the object's displacement rather than distance, which in this case offers the object's shortest path. While the other algorithms seek the distance between two locations using imputed input points and a training sample that isn't relevant to our experiment.

4. 5 Long Short - Term Memory - Stochastic Gradient Descent (LSTM – SGD)

Long Short-Term Memory (LSTM) networks are a form of CNN architecture that was created to overcome the decreasing gradient problem, which is a prevalent issue in CNNs that limits their capacity to acquire long-term relationships in series. LSTMs use a mechanism known as gating to deliberately retain or recall information over some time, making them well-suited for applications involving sequential data, including natural language processing (NLP) as well as time series forecasting. These gates are controlled by sigmoid and tanh activation functions, enabling them to selectively retain and process relevant information over time. Because of

its capacity to efficiently record and analyze complicated temporal associations in data, long short-term memory networks became known as an effective instrument for land subsidence mapping forecasting. Land subsidence, or the slow lowering of the topography of the planet, is a widespread problem in many parts of the world, posing serious threats to facilities ecosystems, and human populations.

$$i_t = \sigma(W_{ii}\, x_t + \, b_{ii} + \, W_{hi} h_t - 1 + \, b_{hi}) \tag{7}$$

$$C_{\sim t} = tanh(W_{ic}\, x_t + \, b_{ic} + \, W_{hc} h_t - 1 + \, b_{hc}) \tag{8}$$

Where W_f is the weight matrix for the forget gate, b_f is the bias vector for the forget gate, h_{t-1} is the hidden state vector from the previous time step, x_t is the input vector at the current time step, σ is the sigmoid activation function.

$$O_t = \sigma(W_{io}\, x_t + \, b_{io} + \, W_{ho} h_t - 1 + \, b_{ho}) \tag{9}$$

These equations show how an LSTM cell processes input at each time step, updates its cell state and produces the next hidden state. These are effective in capturing long-range dependencies in sequential data, making them popular for tasks like natural language processing, speech recognition, and time series prediction.

4.6 Stochastic Gradient Descent (SGD)

Stochastic Gradient Descent (SGD) represents a common optimization technique to identify the shortest path of the cost value in machine learning as well as deep learning. It is very useful when working with huge datasets. Uses one of the chosen at-random training example's gradients from the price function to update the model's parameters. Stochastic Gradient Descent represents a sophisticated optimization approach that handles big datasets effectively and is frequently utilized in machine learning as well as deep learning applications. Its effective execution must adjust hyperparameters including the rate at which it learns. The size of the steps made during parameter updates is determined by the learning rate. These are the fundamental concepts of Stochastic Gradient Descent. Additional approaches and optimizations may be used in practical implementations to increase convergence and stability. The selection of hyperparameters, particularly the learning rate, is critical to the algorithm's performance. The LSTM algorithm is optimized using SGD optimizer which enhances the efficiency of the model is increased, hence accurate of the model determined and performance can be increased with this hybridization of algorithms. Figure 8 shows LSTM with SGD hybrid Model diagram.

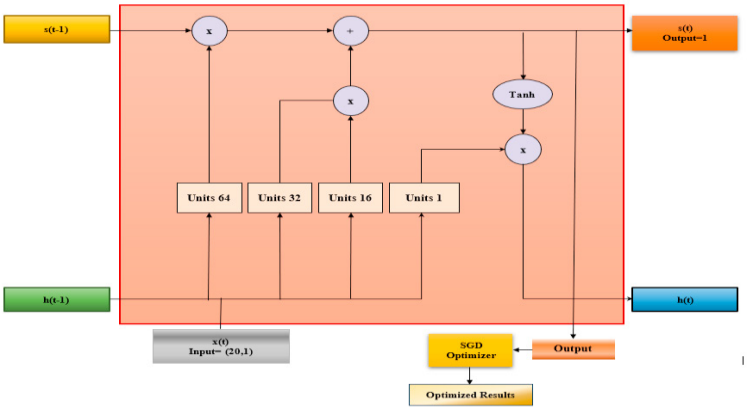


Fig. 8. LSTM with SGD Hybrid Model Diagram.

5. Experiments

5.1 Image Preprocessing

Preprocessing is an important stage for land subsidence prediction because it improves data quality and ready the information for analysis. Missing values were handled by either estimating them or eliminating the

related records. Normalize or standardize numerical properties such that they are all on the same scale. Because of their wider scale, this prevents particular traits from dominating the study. Aggregate temporal information based on the type of the data. Consider collecting data across certain geographical areas to identify local trends if the data has spatial information. The steps involved in the preprocessing steps, enhance the efficiency of the system and are considered as the essential step in the training and validation process. The land subsidence dataset is used for the prediction results determined by implementing the preprocessing techniques.

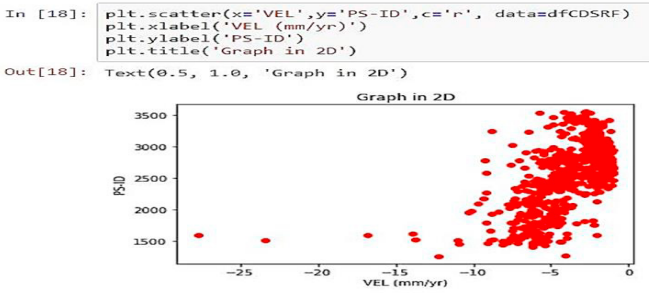


Fig. 9. Outliers Detection

Outlier detection is critical in data pretreatment and analysis. Outliers are data points that deviate significantly from the rest of the data and can have a significant impact on statistical analysis and optimized machine learning models as shown in figure 9.

5.2 Data Augmentation

Augmentation is a machine learning approach that is widely used for an artificially enhance the quantity of a dataset by performing various modifications to the current data. This improves machine learning algorithms' generalization and resilience. Data augmentation is commonly employed in machine learning tasks including image classification, recognizing objects, and division, but its concepts may also be applied to other sorts of data. Augmenting the information assists in preventing overfitting, which is especially important while the initial set of data is small. Augmented information presents the prediction to a broader range of instances, making it more resistant to changes in the input. Figure 10 shows the augmented data of land subsidence mapping using GIS that is employed in the prediction of subsidence.

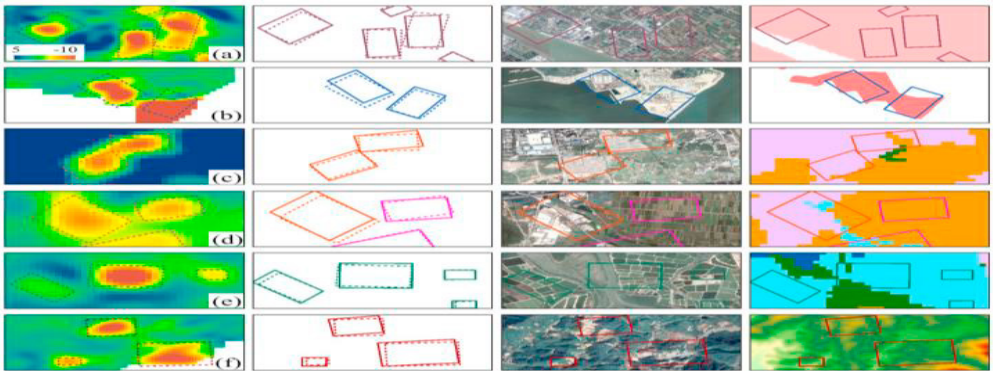


Fig. 10. Land Subsidence Mapping Augmented Data

5.3 Training, Validation, Testing and Accuracy

Several steps are involved in training and verifying a model developed using machine learning with land subsidence prediction to guarantee that the model generalizes successfully to new, unknown information. This research is built on a dataset with relevant attributes and the objective variable. Developing additional characteristics that may improve the model's prediction power. Creating sets for training and validation from the dataset. To teach the model selected, utilize the training dataset. Input characteristics and subsidence data should

be provided to the model. By modifying the internal variables throughout training, the algorithm improves to translate input information to subsidence predictions. Understand the model's predictions by interpreting its learned parameters or using techniques like feature importance analysis.

Table. 2 Training, Testing, Validation and Accuracy Results

Algorithm	Training		Validation		Testing		Training Accuracy (%)
	%	Error	%	Error	%	Error	
ANN	89	0.5	87	0.345	99	7E-45	89
GAN	90	0.2389	100	1.5E-907	99	0.3661	90
KNN	96	0.1485	98	12.893324	100	1.5E-92	96
LSTM - SGD	97	0.7682	94	0.36740	100	0.37718	97

Use measures such as Mean Squared Error (MSE) or any suitable indicators for regression jobs to track learning progress. Utilize techniques like cross-validation to assess how well the model generalizes across different subsets of the training data. Evaluate the model using a new dataset for validation not observed in training. To analyses the systems efficiency performance on the set of validation results, use evaluation measures that include Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), or R-squared. The specifics of these processes will vary depending on the model used, the properties that comprise the dataset, or the aims of the land subsidence forecasting job. To obtain optimal performance, it is critical to properly grasp the issue domain and iteratively modify the model.

5.4 Efficiency and Accuracy

Accuracy is a measure of how effectively a model forecasts the result concerning the real values. It is frequently expressed as a percentage. According to the nature of the task (classification, regression, etc.), measurements like accuracy, precision, recall, F1-score, or the region of the recipient's operating characteristic curve (AUC-ROC). Accuracy may not be the only metric to consider, especially in imbalanced datasets. Precision and recall are often more informative in such cases. Consider the business impact of false positives and false negatives. Some applications may prioritize minimizing one type of error over the other. Fig. 11 and 12 gives the accuracy graph based on training and validation data. The number of accuracies obtained through the hybrid model is shown in Table 2. The ANN algorithm has an accuracy of 89%, GAN has an accuracy of 90%, KNN has an accuracy of 96%, and LSTM + SGD has an accuracy of 97%. By comparing the accuracy obtained using various algorithms, the algorithm LSTM + SGD is considered the most efficient as it yields higher accuracy than other machine learning models.

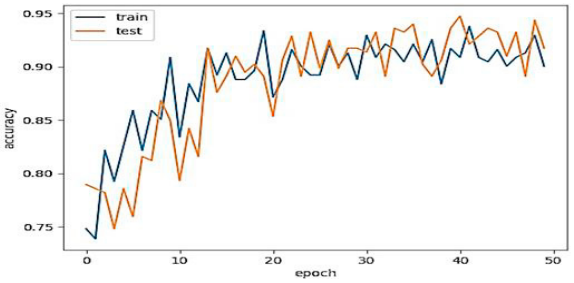


Fig. 11. Model Accuracy is Obtained Using LSTM + SGD

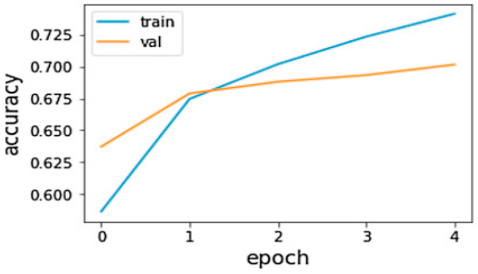


Fig. 12. Accuracy Obtained Using KNN

6. Results and Discussion

The system proposed employs an application of hybrid optimization machine learning models and methodologies efficiency as a hybrid model to anticipate the rising natural disaster land subsidence. The system employs a hybrid model that combines ANN, GAN, KNN, and LSTM + SGD. These are trained on pre-existing picture data from land subsidence mapping using GIS. The model's algorithms aid in comparing and attaining higher efficiency and performance rates. The model's results are compared to the ROC loss value. The number of

accuracies obtained through the hybrid model. The ANN algorithm has an accuracy of 89%, GAN has an accuracy of 90%, KNN has an accuracy of 96%, and LSTM + SGD has an accuracy of 97%. Given these results, the LSTM+SGD is deemed efficient because it outperforms all other algorithms in terms of accuracy.

7. Conclusion and Future work

In this research proposal revealed the efficacy of machine learning optimization algorithms in mapping of land subsidence. The LSTM+SGD network scored the greatest accuracy of 97% among the four algorithms tested, followed by KNN (96%), GAN (90%), and ANN (89%). Because of their capacity to capture complicated nonlinear correlations in data and manage temporal dependencies, LSTM+SGD and KNN models appear to be particularly well-suited for land subsidence prediction. Land subsidence may also cause substantial financial losses by causing structural damage and high maintenance costs for roads, trains, barriers, pipelines, and buildings. GIS allows for the seamless integration of disparate data sources, such as satellite imaging, ground-based measurements, and geological data, into a cohesive geographic framework. This provides a comprehensive perspective of land subsidence patterns and their interactions with other geographical factors. The global bill totals several billions of dollars each year. It will only climb higher in the future as humanity grows and business activity in delta areas intensifies. The future of land deformation monitoring and prediction appears to be in the hands of Artificial Intelligence and Augmented Reality with Deep Learning Techniques.

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